

Shane Schisler

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PROFESSIONAL SUMMARY

Distinguished Software and Security Engineer with 25+ years of experience architecting, building, and scaling complex systems at the intersection of cybersecurity, cloud data platforms, and low-level systems engineering. Proven ability to drive technical strategy from conception to production, influencing C-suite and VP-level stakeholders to invest in forward-looking technologies. Expert in pioneering novel solutions for vulnerability analysis, threat detection, and large-scale data processing that deliver significant business value and competitive advantage.

EXPERIENCE

Contrast Security Inc — *Distinguished Engineer*

APRIL 2018 - PRESENT

Architected and built an agentic security-testing tool in Go that autonomously spins up and exercises arbitrary customer applications. It discovers services, generates Docker/Compose infrastructure, injects the Contrast runtime agent, and drives a multi-turn Claude agent loop until every API endpoint is proven reachable, using a deterministic orchestrator that reserves LLM calls for judgment calls and validates every output artifact before proceeding.

Hardened the agent for production and built its evaluation layer, adding content-hash recipe caching (zero LLM cost on unchanged reruns), per-run USD spend caps, token-budget prompt scheduling and stuck-loop detection, an eval harness benchmarking agent behavior across OWASP vulnerable-application targets, and inference routing through a corporate LLM proxy on AWS Bedrock.

Architected and led the development of a petabyte-scale AWS S3 data lake using Iceberg tables, enabling advanced business analytics and seeding novel AI/ML initiatives. Designed a robust ingestion pipeline with Kafka Connectors and cataloged data with AWS Glue.

Championed and designed a new event-driven, stream-processing architecture using Kafka and Flink. Successfully secured C-suite and VP-level investment in this vision, which decoupled microservices and empowered teams to innovate and deploy faster.

Spearheaded strategic research into eBPF technology, analyzing its potential as both a competitive threat and a new product vector. Focused on its ability to gather application-level context via call stack analysis across Java, Python, Node.js, and Go runtimes.

Directed a strategic partnership with Fraunhofer IEM to productize a novel symbolic static code analysis method, transforming a doctoral dissertation into a commercial product. Led the technical implementation, re-architecting from Kotlin to Rust for enhanced performance.

Developed a Linux kernel module and a more portable user-space solution to automatically inject the Contrast Java Agent, drastically reducing installation friction by 90% and accelerating customer Proof of Value (PoV) sessions.

Designed and prototyped product enhancements to capture inter-service context within microservice architectures, addressing a key weakness in the product's visibility of distributed systems.

Alithix LLC — *Senior Software Engineer*

MARCH 2012 - APRIL 2018

Led a 6-person team in building a custom Type-1 hypervisor from the ground up using C and Intel VT-x, delivering a solution that addressed a critical customer requirement not met by off-the-shelf products.

Authored the winning technical volume for a multi-year, multi-million dollar government contract, directly contributing to the total award by out-scoring the incumbent on their technical volume.

Built a distributed CI testing infrastructure in Python for embedded software, saving an estimated 20

developer hours weekly and providing immediate insight into project health across physical and embedded devices.

Introduced and deployed SaltStack for test infrastructure provisioning, automating the deployment of complex distributed test environments and enabling more frequent, reliable production upgrades to the physical test infrastructure.

Researched and identified APT/malware communication patterns by analyzing netflow and PCAP data with Python data science libraries, informing the development of new detection methods.

Performed low-level firmware and machine code reverse engineering on the x86/AMD64 architectures, and developed interface code for interacting with devices on the Intel southbridge.

G2 Inc — Senior Software Engineer

APRIL 2008 - MARCH 2012

Enabled stealthy malware analysis and hardware reverse engineering by developing hypervisor introspection capabilities, allowing for monitoring of proprietary driver communications.

Automated reverse engineering tasks with IDA Pro scripts, saving hundreds of person-hours annually and accelerating project timelines.

Developed a Linux driver to expose a device's MMIO registers to user-space, enabling developers without kernel experience to conduct research by reverse engineering it in Python, reducing project costs and timelines.

Reverse engineered embedded systems and firmware across x86/AMD64, ST10, and ARM architectures to meet critical customer requirements.

Booz Allen Hamilton — Level III / Associate

JANUARY 2001 - APRIL 2008

Developed a Linux 2.6 kernel driver to re-purpose a standard Atheros wireless NIC for use with custom 802.11 survey software.

Engineered and stress-tested a J2EE / ColdFusion / C++ enterprise web application, identifying and helping resolve key performance bottlenecks.

Conducted 802.11 security research and designed secure Wi-Fi networks utilizing RADIUS and VPN technologies.

TECHNICAL SKILLS

- **Languages:** Rust, Go, C, C++, Java, Kotlin, Python, SQL
- **Cloud & Data Platforms:** AWS (S3, Glue, MSK, RDS, OpenSearch), Kafka, Flink, Apache Iceberg, Spring Boot
- **Security & Analysis:** eBPF, static & dynamic analysis, runtime security analysis, AST manipulation, APT/malware traffic analysis (pcap, netflow)
- **Systems & Low-Level:** Kernel & driver development, hypervisors (VT-x, Xen), binary & firmware reverse engineering (x86/AMD64, ARM), Ghidra, IDA Pro, Wireshark, GDB
- **Data Science & ML:** CNN model tuning (PyTorch, Keras), feature engineering, pandas, NumPy, Kaggle competitions, LLM agent orchestration (Claude CLI/SDK, langchain/langgraph, eval harnesses)

EDUCATION

UMBC, Maryland — B.S. Computer Science

SEPTEMBER 1999 - MAY 2002

Additional graduate coursework in **Computer Architecture** and **Embedded Systems Security** (UMBC, Johns Hopkins University).